Cofinality

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1 Preface

These notes aim to introduce the concept of cofinality, offering the motivations and intuitions behind it. The initial chapters primarily consist of results and examples that explore the idea of cofinality, demonstrating how it naturally arises and examining its key properties.

From there on, almost everything presented can be found in Jech's book - Introduction to Set Theory, all I will do here is a comprehensive and filtered summary, along with some of my own intuitions that may be useful to someone, and some additional comments on things I've found elsewhere.

Notably, the concept of cofinality leads us directly to the Large Cardinal Axioms. We will explore how weakly inaccessible cardinals emerge by addressing a fundamental question in the study of

regular and singular cardinals: does there exist an uncountable regular limit cardinal?

It is important to note that these notes assume the reader has some familiarity with the basics of cardinal and ordinal arithmetic to fully understand the proofs of the theorems. However, even without this background, readers can still grasp the underlying intuition by following the commentary throughout the text, which guides them through a natural exploration of the defined concepts.

2 Motivation

2.1 Classical Examples

Consider the following theorems:

- The finite union of finite sets is finite, i.e., if S is a system of sets such that $|S| < \omega_0$ and, for all $A \in S$, we have $|A| < \omega_0$, then
$$\left|\bigcup S\right| < \omega_0$$
- Similarly, the countable union of countable sets is countable, i.e., if S is a system of sets such that $|S| < \omega_1$ and, for all $A \in S$, we have $|A| < \omega_1$, then
$$\left|\bigcup S\right| < \omega_1$$

In both cases, the key takeaway is that it's difficult to "reach" $\aleph_0$ and $\aleph_1$, respectively, "from below". This phenomenon can be formally understood through the concept of *cofinality*.

To generalize, consider the following question: we know that any limit ordinal is the supremum of the set of all smaller ordinals. In particular, for any limit ordinal λ
$$\lambda = \bigcup \lambda$$
However, we don't necessarily need to take *all* smaller ordinals. It's possible to find a proper subset S of λ such that $\lambda = \sup(S)$. How small can S be? This is the concept we aim to explore.

2.2 Definition of Cofinality

Definition 2.1. We will define $\mathrm{cf}(\alpha)$, the cofinality of α, with α a limit ordinal, as the smallest ordinal ϑ such that α is the limit of an increasing transfinite sequence $\langle \alpha_\nu : \nu < \vartheta \rangle$ of length ϑ, i.e.
$$\alpha = \lim_{\nu \to \vartheta} \alpha_\nu$$

Equivalently, making easier to visualize the relation between this with the discussion above, $\mathrm{cf}(\alpha)$ is the smaller ordinal β such that exists $f : \beta \to \alpha$, satisfying
$$\alpha = \bigcup_{\nu < \beta} f(\nu) = \bigcup \mathrm{Im}(f)$$

i.e., $\mathrm{cf}(\alpha)$ is the smaller ordinal β such that there exists a union of β elements α_ν less than α.

2.3 Singular and Regular Cardinals

We can now define two types of cardinals, κ: those that can be written as a union of κ smaller elements, called singular, and those that cannot, called regular. More formally:

Definition 2.2. An infinite cardinal κ is singular iff there exists an increasing transfinite sequence $\langle \alpha_\nu : \nu < \vartheta \rangle$ of ordinals $\alpha_\nu < \kappa$ where the length ϑ is a limit ordinal less than κ, such that

$$\kappa = \lim_{\nu \to \vartheta} \alpha_\nu$$

An infinite cardinal that is not singular is called regular.

Alternatively, we can define these cardinals more concretely using the following lemma:

Lemma 2.1. An infinite cardinal κ is singular if, and only if, it's the sum of fewer than κ smaller cardinals, i.e.

$$\kappa = \sum_{i \in I} \kappa_i$$

where $|I| < \kappa$ and $\kappa_i < \kappa$, for all $i \in I$.

Proof. ($\Rightarrow$) If κ is singular, then $\kappa = \lim_{\nu \to \vartheta} \alpha_\nu$ for some increasing transfinite sequence with $\alpha_\nu < \kappa$ and $\vartheta < \kappa$. Since all ordinal is the set of all smaller ordinals, then

$$\kappa = \bigcup_{\nu < \vartheta} \alpha_\nu = \bigcup_{\nu < \vartheta} \left(\alpha_\nu \backslash \bigcup_{\xi < \nu} \alpha_\xi \right)$$

defining $A_\nu := \alpha_\nu \backslash \bigcup_{\xi < \nu} \alpha_\xi$, we have that $\langle A_\nu : \nu < \vartheta \rangle$ is a sequence of length less than κ with disjoint sets of cardinality $\kappa_\nu = |A_\nu| = \left| \alpha_\nu \backslash \bigcup_{\xi < \nu} \alpha_\xi \right| \leq |\alpha_\nu| < \kappa$, hence

$$\kappa = \sum_{\nu < \vartheta} \kappa_\nu$$

($\Leftarrow$) Let's assume that $\kappa = \sum_{\alpha < \lambda} \kappa_\alpha$, with $\lambda, \kappa_\alpha < \kappa$, for all $\alpha < \lambda$. Therefore

$$\kappa = \sum_{\alpha < \lambda} \kappa_\alpha = \lambda \cdot \sup_{\alpha < \lambda} \kappa_\alpha$$

and, since $\lambda < \kappa$, then necessarily $\kappa = \sup_{\alpha < \lambda} \kappa_\alpha$, because, otherwise, $\kappa = \lambda$, a contradiction. Then κ is the supremum of the image of the transfinite sequence $\langle \kappa_\alpha : \alpha < \lambda \rangle$, hence, it's easy to find an increasing subsequence with limit κ. ■

Note that such characterization captures well the idea discussed at motivation. We can relate it to the concept of cofinality as follows: it's expected and easy to verify that, if κ is a singular cardinal, then $\mathrm{cf}(\kappa) < \kappa$, i.e., there is an increasing sequence of length smaller than κ with elements smaller than κ, but which has κ as a limit and, if κ is regular, since $\mathrm{cf}(\kappa) \leq \kappa$, we have that $\mathrm{cf}(\kappa) = \kappa$.

3 Interesting Properties and Examples

3.1 Distribution of Regular and Singular Cardinals

To facilitate notation, we will say that a cardinal $\aleph_\alpha$ is a *successor cardinal* if α is a successor ordinal, and we will say that it's a *limit cardinal* if α is a limit ordinal.

Let's explore the concept we create with some cardinals. For the cases used in the motivation, we have that $\mathrm{cf}(\aleph_0) = \aleph_0$ and $\mathrm{cf}(\aleph_1) = \aleph_1$, i.e., both are regular cardinals, since they can't be written as a union of $\aleph_0$ or $\aleph_1$ smaller elements, respectively. In particular, the following theorem will prove that every successor cardinal $\aleph_{\alpha+1}$ is a regular cardinal, that is, it's not easy to "reach them from below", only with smaller sets.

Theorem 3.1. Every successor cardinal $\aleph_{\alpha+1}$ is a regular cardinal.

Proof. Assume by contradiction that $\aleph_{\alpha+1}$ is singular, by Lemma 2.1, it can be written as a sum of a smaller number of smaller cardinals:

$$\aleph_{\alpha+1} = \sum_{i \in I} \kappa_i$$

with $|I| < \aleph_{\alpha+1}$ and $\kappa_i < \aleph_{\alpha+1}$, $\forall i \in I$. Therefore $|I|, \kappa_i \leq \aleph_\alpha$, $\forall i \in I$, so, we have

$$\aleph_{\alpha+1} = \sum_{i \in I} \kappa_i \leq \sum_{i \in I} \aleph_\alpha = \aleph_\alpha \cdot |I| \leq \aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$$

a contradiction, then $\aleph_{\alpha+1}$ is a regular cardinal. ■

On the other hand, we have

$$\begin{aligned}
\aleph_\omega &= \lim_{n \to \omega} \aleph_n \\
\aleph_{\omega+\omega} &= \lim_{n \to \omega} \aleph_{\omega+n} \\
\aleph_{\omega \cdot \omega} &= \lim_{n \to \omega} \aleph_{\omega \cdot n} \\
\aleph_{\omega_1} &= \lim_{\alpha \to \omega_1} \aleph_\alpha
\end{aligned}$$

It's easy to see that the vast majority of limit cardinals we have contacts with, except for $\aleph_0$, are indeed singular; we will state this more precisely in the next theorem. Furthermore, each cardinal on the examples shown above are singular, the first three had cofinality ω and the last one cardinality ω_1. We will see how powerful this is when we present other intuitive ways of visualizing them, but for now let us answer some natural questions that arise, such as: are there arbitrarily large singular cardinals? Or rather, are all uncountable limit cardinals singular?

3.2 Some Basic Properties

Let's us first show that there do indeed exist arbitrarily large singular cardinals, and further, if $\aleph_\alpha$ is a singular limit cardinal, then the smallest limit cardinal larger than $\aleph_\alpha$, i.e., the next limit cardinal, is also singular.

Lemma 3.1. There are arbitrarily large singular cardinals.

Proof. Let $\aleph_\alpha$ be an arbitrary cardinal and consider the following sequence

$$\aleph_\alpha, \aleph_{\alpha+1}, \aleph_{\alpha+2}, \ldots, \aleph_{\alpha+n}, \ldots$$

For $n \in \omega$. Therefore

$$\aleph_{\alpha+\omega} = \lim_{n\to\omega} \aleph_{\alpha+n}$$

and, then, $\aleph_{\alpha+\omega}$ is a singular cardinal greater than $\aleph_\alpha$. ■

As a scholium of the previous lemma, we have

Corollary 3.1. If $\aleph_\alpha$ is a singular cardinal, then the next limit cardinal $\aleph_{\alpha+\omega}$ is also singular. In particular, if $\aleph_\alpha$ was regular, the next limit cardinal is also singular.

With this, we can have a visualization of which of the first cardinals are regular or singular. We have that, after $\aleph_0$, each successor cardinal below $\aleph_\omega$ is regular, and $\aleph_\omega$ is itself singular. We can repeat it from $\aleph_\omega$ to $\aleph_{\omega\cdot 2}$, $\aleph_{\omega^\omega}$, $\aleph_{\varepsilon_0}$, and so on. In fact, even reaching $\aleph_{\omega_1}$, which is singular, we will have the same between it and the next limit $\aleph_{\omega_1+\omega}$. So the "line" of cardinals has, apparently, almost always a singular cardinal in the limit points, and always countable many regular successor cardinals between them.

3.3 Uncountable Limit Regular Cardinals

Let's now investigate how would be an uncountable limit regular cardinal. Let $\aleph_\lambda$ be such cardinal, since λ is a limit ordinal, then

$$\aleph_\lambda = \lim_{\alpha\to\lambda} \aleph_\alpha$$

i.e., $\aleph_\lambda$ is the limit of an increasing sequence of length α, in particular, we could change such sequence to another one of length $\mathrm{cf}(\lambda)$. It's easy to see how this imply the next result

Lemma 3.2. If λ is a limit ordinal, then $\mathrm{cf}(\aleph_\lambda) = \mathrm{cf}(\lambda)$

Now, since in our case $\aleph_\lambda$ is also a regular cardinal, then $\aleph_\lambda = \mathrm{cf}(\aleph_\lambda) = \mathrm{cf}(\lambda) \leq \lambda$, which imply

$$\boxed{\aleph_\lambda = \lambda}$$

Such cardinals are called aleph fixed points, since, well, they're a fixed point of the aleph operation.

Now, we already know $\aleph_\omega \neq \omega$, and $\aleph_{\omega_1} \neq \omega_1$, $\aleph_{\omega_2} \neq \omega_2$, etc. and the gap between the indices ω_1 and 1, ω_2 and 2, and so on keep increasing! Similarly, we have $\aleph_{\aleph_\omega} \neq \aleph_\omega$. Repeating this arguments it's easy to see that such finite towers of $\aleph$'s cannot reach an aleph fixed point. In fact, it shows that the first aleph fixed point is of the form

$$\aleph_{\aleph_{\aleph_{\ddots}}} = \bigcup_{n\in\omega} \aleph_{\aleph_{\aleph_{\ddots_{\aleph_0}}}}$$

where we have n $\aleph$'s on the RHS.

Such property and example suggest these cardinals should be really big. However, the next Lemma reveals us that, in fact, this property isn't too strong as it seems, since we can actually prove that there are arbitrarily large singular aleph fixed points, as the next lemma[1] states.

footnote hyperlink

Lemma 3.3. There are arbitrarily large singular aleph fixed points.

Proof. Let κ be an arbitrary cardinal and consider the following sequence α_n defined recursively as $\alpha_0 = \aleph_\gamma$, $\alpha_{n+1} = \aleph_{\alpha_n}$, for each $n \in \omega$. Now, assume by contradiction that there is not singular $\alpha \geq \kappa$ which is a aleph fixed point, then α_n is an increasing sequence; let $\alpha := \lim_{n\to\omega} \alpha_n$. Obviously the sequence $\aleph_{\alpha_n} = \alpha_{n+1}$ also has limit $\aleph_\alpha$, therefore, we have that

$$\aleph_\alpha = \lim_{n\to\omega} \aleph_{\alpha_n} = \lim_{n\to\omega} \alpha_{n+1} = \alpha$$

Then $\alpha = \aleph_\alpha$ is a fixed aleph point which is also a limit of an increasing sequence of length ω, i.e., $\mathrm{cf}(\alpha) = \omega < \alpha$, then it's singular. ■

The next natural step is then ask what would be an example of an uncountable regular limit cardinal. Well, surprisingly, such cardinals, named weakly inaccessible cardinals, are so big that ZFC can't prove they exist! Inaccessible cardinals are some of the first cardinals in a larger class of properties named Large Cardinals Properties.

Unfortunatelly, the proof is not that easy, since we need to use a consistent result, namely, that GCH is consistent with ZFC[2]. However, we can prove it for a somewhat similar cardinal, named strongly limit cardinals, you can see how to prove it in the Large Cardinals notes, and then how to use it to show ZFC also can't prove weakly inaccessible exist.

add hyperlink

3.4 Rambling on Cofinality

Now we already motivated Large Cardinal, let's ramble a little about cofinality. Some books, before defining cofinality, use the following definitions: a set $C \subseteq \lambda$ is cofinal or unbounded in λ if for all $\alpha < \lambda$, there exists a $\gamma \in C$ such that $\alpha < \gamma$. It's not hard to see that this is equivalent to $\bigcup C = \lambda$.

We can then define $\mathrm{cf}(\lambda)$ as the least cardinality of an unbounded subset of λ. In other words, what we are saying is that subsets of regular cardinal need almost all elements to become unbounded in it, whereas singular cardinals don't.

Now, imagine we have some property about ordinals smaller than $\aleph_\omega$, and such property is somewhat inductive. Then the process of extending this property to $\aleph_\omega$ itself would require only ω many steps, since $\mathrm{cf}(\aleph_\omega) = \omega$, whereas, for example, doing the same for $\aleph_1$ would require uncountable many steps.

Of course none of this is really formal, but you will get your own intuition about this while getting deeper and deeper in set theory. You will see that a lot of theorems states something only for regular or only for singular cardinals, and that they influence a lot, for example, in cardinal arithmetic. In

[1] In fact, this lemma is the toy version of a more general fact known as The Fixed-Point Lemma for normal functions, which guarantees us that every normal function has arbitrarily large fixed points. You can see this in its full generality on my Large Cardinals notes, the proof follows the same idea as the one used to prove the lemma.

[2] You can find a proof of it in my Consistency Results notes, in Section 3. The Constructible Universe L.

fact, we will see in the next chapters that cardinal arithmetic depends a lot of when the cardinals we are using are either singular or regular.

3.5 Additional Properties of Cofinality

Let's finish this section with some extra results about cofinality.

We already proved that if $\aleph_\alpha$ is singular, then $\mathrm{cf}(\omega_\alpha) < \omega_\alpha$ and, if it's regular, then $\mathrm{cf}(\omega_\alpha) = \omega_\alpha$. We will now show that

Lemma 3.4. If a limit ordinal α isn't a cardinal, then $\mathrm{cf}(\alpha) < \alpha$. In particular, for all limit ordinal α, we have $\mathrm{cf}(\alpha) = \alpha$ if, and only if, α is a regular cardinal.

Proof. Let α be an ordinal which is not a cardinal, then $\kappa = |\alpha| < \alpha$ and there is a bijection $f : \kappa \to \alpha$, i.e., a sequence $\langle \alpha_\nu = f(\nu) : \nu < \kappa \rangle$ of length κ such that $\{\alpha_\nu : \nu < \kappa\} = \alpha$.

Now, even if such sequence was not increasing, we could find, using transfinite induction, an increasing subsequence with limit α. And, since the length of the subsequence is at most $\kappa < \alpha$, then $\mathrm{cf}(\alpha) < \alpha$. ■

Lemma 3.5. For all limit ordinal α, we have

$$\mathrm{cf}(\mathrm{cf}(\alpha)) = \mathrm{cf}(\alpha)$$

Proof. Let $\vartheta = \mathrm{cf}(\alpha)$. Clearly ϑ is a limit ordinal and we already know that $\mathrm{cf}(\vartheta) \leq \vartheta$. Now, assume by a contradiction that $\gamma = \mathrm{cf}(\vartheta) < \vartheta$, then there is an increasing sequence $\langle \nu_\xi : \xi < \gamma \rangle$ with $\lim_{\xi \to \gamma} \nu_\xi = \vartheta$. But since $\vartheta = \mathrm{cf}(\alpha)$, there is also a $\langle \alpha_{\nu_\xi} : \xi < \gamma \rangle$ of length $\gamma < \vartheta$ with limit α, contradicting the minimality of ϑ. ■

Corollary 3.2. For all limit ordinal α, $\mathrm{cf}(\alpha)$ is a regular cardinal.

4 Cardinals Exponentiation

While addition and multiplication of cardinals is very simple, since

$$\aleph_\alpha + \aleph_\beta = \aleph_\alpha \cdot \aleph_\beta = \max\{\aleph_\alpha, \aleph_\beta\}$$

exponentiation is way more complex.

In particular, we will see that The Generalized Continuum Hypothesis (GCH):

$$2^{\aleph_\alpha} = \aleph_{\alpha_+1}, \quad \text{for all } \alpha$$

simplify cardinal exponentiation enormously and, without it, for regular cardinals there isn't much to prove[3] besides

$$2^{\aleph_\alpha} \geq \aleph_{\alpha+1}, \qquad \alpha \leq \beta \Rightarrow 2^{\aleph_\alpha} \leq 2^{\aleph_\beta} \tag{1}$$

and the next result about cofinality.

4.1 König's Theorem

Before proving some theorems in cardinal exponentiation, let's prove one of the most famous results in cardinal arithmetic, known as König's Theorem, which is a strict inequality! Notice how rare this is, the only other strict inequality we saw in cardinal arithmetic was probably Cantor's Theorem. In fact, we will see Cantor is a direct consequence of König's.

Theorem 4.1. (**König's Theorem**) If $\kappa_i < \lambda_i$, where κ_i and λ_i are cardinal, $\forall i \in I$, then

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i$$

The idea behind the proof is: If $\kappa_i < \lambda_i$, then every function $f_i : \kappa_i \to \lambda_i$ isn't surjective, therefore, we can find an element $b_i \in \lambda_i \backslash \mathrm{Im}(f_i)$, and then we obtain, for each function $f : \bigcup_{i \in I} A_i \to \prod_{i \in I} B_i$, the element $b = \langle b_i : i \in I \rangle$ which is not in the image of f.

Proof. Let $\langle A_i : i \in I \rangle$ and $\langle B_i : i \in I \rangle$ be such that $|A_i| = \kappa_i$ and $|B_i| = \lambda_i$, with A_i mutually disjoint, $\forall i \in I$. Since $|A_i| < |B_i|$, then, given $f_i : A_i \to B_i$ injective, we know they can't be injective. So, define

$$B_i^\dagger = \{x \in B_i : x \notin f_i[A_i]\}$$

Since f_i isn't surjective, $B_i^\dagger \neq \emptyset$, $\forall i \in I$, then, by AC, we can choose $b_i^\dagger \in B_i^\dagger$, $\forall i \in I$. Now, let

$$F : \bigcup_{i \in I} A_i \to \prod_{i \in I} B_i$$

be the unique function satisfying

$$\pi_j(F(a)) = \begin{cases} f_i(a), & i = j \\ b_j^\dagger, & i \neq j \end{cases}$$

Now, if $F(x) = F(y)$, with $x \in A_i$ and $y \in A_j$, then

$$\pi_i(F(x)) = \pi_i(F(y))$$

if $i \neq j$, then

$$\pi_i(F(x)) = f_i(x) = b_i^\dagger = \pi_i(F(y))$$

a contradiction, since $b_i^\dagger \notin f_i[A_i]$. If $i = j$, then $f_i(x) = f_i(y)$, but, since f_i is injective, $x = y$. This proves F is an injection. Let's now prove there is no surjection.

[3] in fact, this is a statement in the formal sense. We have a lot of results of the form: giving some relations between cardinals, any possible assignment to those cardinals that doesn't contradicts such relations is realized in some model.

Assume by contradiction that there is some $G : \bigcup_{i\in I} A_i \to \prod_{i\in I} B_i$ surjective, and define

$$B_i^\ddagger = \{x \in B_i : x \notin \pi_i(G[A_i])\}$$

Since $(\pi_i \circ G) : A_i \to B_i$, then it's not surjective and, therefore, $B_i^\ddagger \neq \emptyset$, $\forall i \in I$. By AC there is a $b = \left\langle b_i^\ddagger : i \in I \right\rangle$ such that $b_i^\ddagger \in B_i^\ddagger$. Since G is surjective, there is a $x \in A_i$, for some $i \in I$, such that $G(x) = b^\ddagger$, so $\pi_i(G(x)) = \pi_i\left(b^\ddagger\right) \in B_i^\ddagger$, a contradiction, then G is not surjective. ■

Notice that the choices of $B_i^\dagger$ and $B_i^\ddagger$ in the proof looks like the choice of $X = \{x \in A : x \notin f(x)\}$ in Cantor's Theorem proof and, in fact, Cantor's Theorem is a corollary of König's Theorem for $\kappa_i = 1$ and $\lambda_i = 2$, $\forall i \in I$.

4.1.1 Cardinal Exponentiation for Regular Cardinals

With such big results in hands, we can prove the following lemma

Lemma 4.1. For any ordinal α

$$\mathrm{cf}\left(2^{\aleph_\alpha}\right) > \aleph_\alpha$$

Note that this lemma restricts the possible values $2^{\aleph_\alpha}$ can assume, depending on its cofinality. As an example, which also reinforce the comments in subsection 3.4 and in the last footnote, notice that we can't have $2^{\aleph_0} = \aleph_\omega$, since then we would have $\aleph_0 = \mathrm{cf}\,(\aleph_\omega) = \mathrm{cf}\left(2^{\aleph_0}\right) > \aleph_0$, a contradiction.

In fact, Robert Solovay published a result in a really famous one page paper named "$2^{\aleph_0}$ can be anything it ought to be" which basically says that the only two things we can prove about the continuum is: that it's uncountable, and that it has uncountable cofinality. Formally, the result says that for any uncountable cardinal κ with uncountable cofinality, there is a model of ZFC where this is realized, i.e., where $2^{\aleph_0} = \kappa$.

Nowadays, such result is a direct consequence of a more general one known as Easton's Theorem, that says basically the same thing, but for any regular cardinal, i.e., the only constraints on permissible values for 2^κ when κ is a regular cardinal are (1) and the previous lemma.

Proof. Let $\vartheta = \mathrm{cf}\left(2^{\aleph_\alpha}\right)$, then, there are $\kappa_\nu < 2^{\aleph_\alpha}$, $\nu < \vartheta$ such that

$$2^{\aleph_\alpha} = \sum_{\nu<\vartheta} \kappa_\nu$$

Using König's Theorem for $\lambda_\nu = 2^{\aleph_\alpha}$, we have

$$2^{\aleph_\alpha} = \sum_{\nu<\vartheta} \kappa_\nu < \prod_{\nu<\vartheta} 2^{\aleph_\alpha} = \left(2^{\aleph_\alpha}\right)^\vartheta$$

Now, assume by contradiction that $\vartheta \leq \aleph_\alpha$, then

$$2^{\aleph_\alpha} > \left(2^{\aleph_\alpha}\right)^\vartheta \leq \left(2^{\aleph_\alpha}\right)^{\aleph_\alpha} = 2^{\aleph_\alpha \cdot \aleph_\alpha} = 2^{\aleph_\alpha}$$

a contradiction, so $\vartheta = \mathrm{cf}\left(2^{\aleph_\alpha}\right) > \aleph_\alpha$. ■

Notice how we didn't use anything special about 2, and then we can repeat the exact same argument for any cardinal $\kappa > 1$, giving us:

Corollary 4.1. For any cardinals $\kappa, \lambda > 1$, we have

$$\operatorname{cf}\left(\kappa^{\lambda}\right) > \lambda$$

4.1.2 Digression on Exponentiation for Singular Cardinals

As we saw, the inequalities above are the only ones we can prove about regular cardinals, the next natural step is to ask if there is a generalization, or some similar result about singular cardinals. In fact, for singular cardinals κ, various additional constraints in the behavior of 2^{κ} are known.

By know, the most interesting one is Silver's Theorem, since it implies that Easton's Theorem can't be extended to include singular cardinals. But, nowadays, Shelah's work on PCF Theory gives a lot of results on the possible values of 2^{κ}, for κ singular. It shows that such values are strongly influenced by the values on smaller cardinals, as we will see next and when proving Silver's Theorem, whereas Easton shows this is not the case for regular cardinals.

Theorem 4.2. Let $\aleph_{\alpha}$ be a singular cardinal, if $2^{\aleph_{\xi}} = \aleph_{\beta}$ for all $\xi < \alpha$, then $2^{\aleph_{\alpha}} = \aleph_{\beta}$.

Proof. Since $\aleph_{\alpha}$ is singular, there are $\kappa_i < \aleph_{\alpha}$, $i \in I$, with $|I| = \aleph_{\lambda} < \aleph_{\alpha}$ such that

$$2^{\aleph_{\alpha}} = \sum_{i \in I} \kappa_i$$

By hypothesis $2^{\kappa_i} = 2^{\aleph_{\lambda}} = \aleph_{\beta}$, then

$$2^{\aleph_{\alpha}} = 2^{\sum_{i \in I} \kappa_i} = \prod_{i \in I} 2^{\kappa_i} = \prod_{i \in I} \aleph_{\beta} = (\aleph_{\beta})^{\aleph_{\lambda}} = \left(2^{\aleph_{\lambda}}\right)^{\aleph_{\lambda}} = 2^{\aleph_{\lambda}} = \aleph_{\beta}$$

■

4.2 Evaluating $\aleph_{\alpha}^{\aleph_{\beta}}$

Let's first state some important lemmas. The first simplify $\aleph_{\alpha}^{\aleph_{\beta}}$ for when $\alpha \leq \beta$.

Lemma 4.2. If $\alpha \leq \beta$, then

$$\aleph_{\alpha}^{\aleph_{\beta}} = 2^{\aleph_{\beta}}$$

Proof. Since $2 < \aleph_{\alpha}$, we have $2^{\aleph_{\beta}} \leq \aleph_{\alpha}^{\aleph_{\beta}}$. By Cantor's Theorem $\aleph_{\alpha} < 2^{\aleph_{\alpha}}$, then

$$\aleph_{\alpha}{}^{\aleph_{\beta}} \leq \left(2^{\aleph_{\alpha}}\right)^{\aleph_{\beta}} = 2^{\aleph_{\alpha} \cdot \aleph_{\beta}} = 2^{\aleph_{\beta}}$$

■

The next natural step is trying to calculate $\aleph_\alpha^{\aleph_\beta}$ for when $\alpha > \beta$. The next lemma will show itself useful when trying to do this.

Lemma 4.3. If $\alpha \geq \beta$, then

$$\left|[\omega_\alpha]^{\aleph_\beta}\right| = \aleph_\alpha^{\aleph_\beta}$$

Proof. Since $|\omega_\alpha \times \omega_\beta| = \aleph_\alpha \cdot \aleph_\beta = \aleph_\alpha = |\omega_\alpha|$, then $\left|[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}\right| = \left|[\omega_\alpha]^{\aleph_\beta}\right|$. Now, every function $f : \omega_\beta \to \omega_\alpha$ is in $[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}$, therefore $\omega_\alpha^{\omega_\beta} \subseteq [\omega_\alpha \times \omega_\beta]^{\aleph_\beta}$ and so

$$\left|\omega_\alpha^{\omega_\beta}\right| = \aleph_\alpha^{\aleph_\beta} \leq \left|[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}\right| = \left|[\omega_\alpha]^{\aleph_\beta}\right|$$

For the reverse inequality, if $X \in [\omega_\alpha]^{\aleph_\beta}$, then there is some f in ω_b such that $\mathrm{Im}(f) = X$. With this, given a function f_X for each $X \in [\omega_\alpha]^{\aleph_\beta}$, let $F(X) = f_X$. If $X \neq Y$, then $\mathrm{Im}(f_X) = X$ and $\mathrm{Im}(f_Y) = Y$, so $F(X) = f_X \neq f_Y = F(y)$, i.e., F is injective, then

$$\left|[\omega_\alpha]^{\aleph_\beta}\right| \leq \left|\omega_\alpha^{\omega_\beta}\right| = \aleph_\alpha^{\aleph_\beta}$$

■

4.2.1 Under GCH

Now, as a warmup, let's see what we got under the Generalized Continuum Hypothesis (GCH).

Theorem 4.3. (GCH) If $\aleph_\alpha$ is a regular cardinal, then

$$\aleph_\alpha^{\aleph_\beta} = \begin{cases} \aleph_\alpha, & \text{if } \alpha > \beta \\ \aleph_{\beta+1}, & \text{if } \alpha \leq \beta \end{cases}$$

Proof. If $\alpha \leq \beta$, by Lemma 4.2 $\aleph_\alpha^{\aleph_\beta} = 2^{\aleph_\beta} = \aleph_{\beta+1}$. Let then $\alpha > \beta$. If we let $S = [\omega_\alpha]^{\aleph_\beta}$, by Lemma 4.3 we have $|S| = \aleph_\alpha^{\aleph_\beta}$. In subsection 3.4, we saw $\mathrm{cf}(\kappa)$ can be interpreted as the least size of an unbounded set in κ. In particular, since $\mathrm{cf}(\aleph_\alpha) = \aleph_\alpha$ for regular cardinals, we know every $X \subseteq \omega_\alpha$, with $|X| < \aleph_\alpha$, is limited. Let then

$$B = \bigcup_{\delta < \omega_\alpha} \mathcal{P}(\delta)$$

be the collection of all limited subsets of ω_α. We claim $|B| \leq \aleph_\alpha$; then, since $S \subset B$, we have $|S| = \aleph_\alpha^{\aleph_\beta} = \aleph_\alpha$, since clearly $|S| = \aleph_\alpha^{\aleph_\beta} \geq \aleph_\alpha$. From the definition of B we have

$$|B| \leq \sum_{\delta < \omega_\alpha} 2^{|\delta|}$$

However, for any cardinal $\aleph_\gamma < \aleph_\alpha$, we also have $2^{\aleph_\gamma} = \aleph_{\gamma+1} \leq \aleph_\alpha$, therefore $2^{|\delta|} \leq \aleph_\alpha$, for any $\delta < \omega_\alpha$, then

$$|B| \leq \sum_{\delta<\omega_\alpha} 2^{|\delta|} \leq \sum_{\delta<\omega\alpha} \aleph_\alpha = \aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$$

■

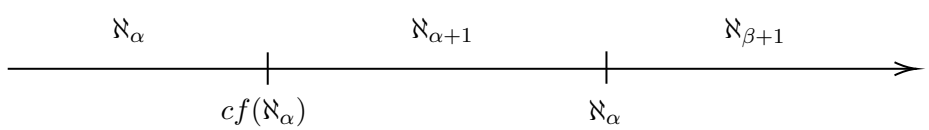
Let's now prove a similar, but more complicated, formula for singular cardinals.

Theorem 4.4. (GCH) If $\aleph_\alpha$ is a singular cardinal, then

$$\aleph_\alpha^{\aleph_\beta} = \begin{cases} \aleph_\alpha, \text{ if } \aleph_\beta < \mathrm{cf}\,(\aleph_\alpha), \\ \aleph_{\alpha+1}, \text{ if } \mathrm{cf}\,(\aleph_\alpha) \leq \aleph_\beta \leq \aleph_\alpha, \\ \aleph_{\beta+1}, \text{ if } \aleph_\beta \geq \aleph_\alpha \end{cases}$$

Schematically, we obtain

$$f(\aleph_\beta) = \aleph_\alpha^{\aleph_\beta}$$



Proof. If $\beta \geq \alpha$, then $\aleph_\beta \geq \aleph_\alpha$ and $\aleph_\alpha^{\aleph_\beta} = 2^{\aleph_\beta} = \aleph_{\beta+1}$. If $\aleph_\beta < \mathrm{cf}\,(\aleph_\alpha)$, then any $X \subseteq \omega_\alpha$ with $|X| = \aleph_\beta < \mathrm{cf}\,(\aleph_\alpha) < \aleph_\alpha$ is bounded in ω_α and, therefore, by the same argument as in the previous theorem, we have $\aleph_\alpha^{\aleph_\beta} = \aleph_\alpha$. Let then $\mathrm{cf}\,(\aleph_\alpha) \leq \aleph_\beta \leq \aleph_\alpha$. On the one hand

$$\aleph_\alpha \leq \aleph_\alpha^{\aleph_\beta} \leq \aleph_\alpha^{\aleph_\alpha} = 2^{\aleph_\alpha} = \aleph_{\alpha+1} \tag{†}$$

But on the other, by Corollary 4.1, $\mathrm{cf}\left(\aleph_\alpha^{\aleph_\beta}\right) > \aleph_\beta$ and, since $\aleph_\beta \geq \mathrm{cf}\,(\aleph_\alpha)$, then $\mathrm{cf}\left(\aleph_\alpha^{\aleph_\beta}\right) \neq \mathrm{cf}\,(\aleph_\alpha)$, so $\aleph_\alpha^{\aleph_\beta} \neq \aleph_\alpha$. With this and (†), the only possible value for $\aleph_\alpha^{\aleph_\beta}$ is $\aleph_{\alpha+1}$. ■

4.2.2 Hausdorff's Formula

Lastly, let's see what we can do without GCH.

Theorem 4.5. (Hausdorff's Formula) For any ordinals α and β, we have

$$\aleph_{\alpha+1}^{\aleph_\beta} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

Proof. If $\beta \geq \alpha + 1$, then, since $\aleph_{\alpha+1} \leq \aleph_\beta \leq 2^{\aleph_\beta}$, we have

$$\aleph_{\alpha+1}^{\aleph_\beta} = 2^{\aleph_\beta} = 2^{\aleph_\beta} \cdot \aleph_{\alpha+1} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

Let then $\beta \leq \alpha$. Since $\aleph_\alpha^{\aleph_\beta} \leq \aleph_{\alpha+1}^{\aleph_\beta}$ and $\aleph_{\alpha+1} \leq \aleph_{\alpha+1}^{\aleph_\beta}$, then

$$\aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1} \leq \aleph_{\alpha+1}^{\aleph_\beta} \cdot \aleph_{\alpha+1}^{\aleph_\beta} = \aleph_{\alpha+1}^{\aleph_\beta}$$

Therefore, it's sufficient to show $\aleph_{\alpha+1}^{\aleph_\beta} \leq \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$. Since $\omega_\beta < \omega_{\alpha+1}$ and $\omega_{\alpha+1}$ is regular, then for any $f : \omega_\beta \to \omega_{\alpha+1}$, f is bounded, i.e., there is some $\gamma < \omega_{\alpha+1}$ such that $f(\xi) < \gamma$, for every $\xi < \omega_\beta$. With this

$$\omega_{\alpha+1}^{\omega_\beta} = \bigcup_{\gamma < \omega_{\alpha+1}} \gamma^{\omega_\beta}$$

Now, any $\gamma < \omega_{\alpha+1}$ is such that $|\gamma| \leq \aleph_\alpha$ and, since

$$\left| \bigcup_{\gamma < \omega_{\alpha+1}} \gamma^{\omega_\beta} \right| \leq \sum_{\gamma < \omega_{\alpha+1}} |\gamma|^{\aleph_\beta}$$

then

$$\aleph_{\alpha+1}^{\aleph_\beta} \leq \sum_{\gamma < \omega_{\alpha+1}} |\gamma|^{\aleph_\beta} \leq \sum_{\gamma < \omega_{\alpha+1}} \aleph_\alpha^{\aleph_\beta} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

■

5 Silver's Theorem

todo